Homework 5

Problem 1 Flapping angle $\phi(t)$ (controlled input), passive pitching angle is: $q=\psi$, $\dot{q}=\dot{\psi}$

Body angular velocity components $w_x=\dot{\psi}$, $w_y=\dot{\phi}\cos\psi$, $w_z=-\dot{\phi}\sin\psi$ $\therefore$ $w=\begin{bmatrix}\dot{\psi}\\ \dot{\phi}\cos\psi\\ -\dot{\phi}\sin\psi\end{bmatrix}$

Inertia matrix: $I=\begin{bmatrix}I_{xx} & -I_{xy} & 0\\ -I_{xy} & I_{yy} & 0\\ 0 & 0 & I_{xx}+I_{yy}\end{bmatrix}$, Rotational kinetic energy $T=\frac{1}{2}w^T I w$

$$Iw=\begin{bmatrix}I_{xx} & -I_{xy} & 0\\ -I_{xy} & I_{yy} & 0\\ 0 & 0 & I_{xx}+I_{yy}\end{bmatrix}\begin{bmatrix}\dot{\psi}\\ \dot{\phi}\cos\psi\\ -\dot{\phi}\sin\psi\end{bmatrix}=\begin{bmatrix}I_{xx}(\dot{\psi})+(-I_{xy})(\dot{\phi}\cos\psi)+0\\ (-I_{xy})(\dot{\psi})+I_{yy}(\dot{\phi}\cos\psi)+0\\ 0+0+(I_{xx}+I_{yy})(-\dot{\phi}\sin\psi)\end{bmatrix}=\begin{bmatrix}I_{xx}\dot{\psi}-I_{xy}\dot{\phi}\cos\psi\\ -I_{xy}\dot{\psi}+I_{yy}\dot{\phi}\cos\psi\\ -(I_{xx}+I_{yy})\dot{\phi}\sin\psi\end{bmatrix}$$

$$w^TIw=\begin{bmatrix}\dot{\psi} & \dot{\phi}\cos\psi & -\dot{\phi}\sin\psi\end{bmatrix}\begin{bmatrix}I_{xx}\dot{\psi}-I_{xy}\dot{\phi}\cos\psi\\ -I_{xy}\dot{\psi}+I_{yy}\dot{\phi}\cos\psi\\ -(I_{xx}+I_{yy})\dot{\phi}\sin\psi\end{bmatrix}=I_{xx}\dot{\psi}^2-\underline{I_{xy}\dot{\psi}\dot{\phi}\cos\psi}-\underline{I_{xy}\dot{\phi}\dot{\psi}\cos\psi}+I_{yy}\dot{\phi}^2\cos^2\psi+(I_{xx}+I_{yy})\dot{\phi}^2\sin^2\psi$$

$$=I_{xx}\dot{\psi}^2-2I_{xy}\dot{\phi}\dot{\psi}\cos\psi+\underline{I_{yy}\dot{\phi}^2\cos^2\psi}+I_{xx}\dot{\phi}^2\sin^2\psi+\underline{I_{yy}\dot{\phi}^2\sin^2\psi}\qquad \cos^2\psi+\sin^2\psi=1$$

$$=I_{xx}\dot{\psi}^2-2I_{xy}\dot{\phi}\dot{\psi}\cos\psi+I_{yy}\dot{\phi}^2+I_{xx}\dot{\phi}^2\sin^2\psi$$

$T=\frac{1}{2}w^TIw=\frac{1}{2}I_{xx}\dot{\psi}^2-I_{xy}\dot{\psi}\dot{\phi}\cos\psi+\frac{1}{2}I_{yy}\dot{\phi}^2+\frac{1}{2}I_{xx}\dot{\phi}^2\sin^2\psi$ Euler-Lagrange equation $\frac{d}{dt}\left(\frac{\partial T}{\partial\dot{\psi}}\right)-\frac{\partial T}{\partial\psi}=M_x$

$\frac{\partial T}{\partial\dot{\psi}}=I_{xx}\dot{\psi}-I_{xy}\dot{\phi}\cos\psi$ $\frac{d}{dt}\left(\frac{\partial T}{\partial\dot{\psi}}\right)=\frac{d}{dt}\left(I_{xx}\dot{\psi}-I_{xy}\dot{\phi}\cos\psi\right)=I_{xx}\ddot{\psi}+I_{xy}\dot{\phi}\sin\psi\,\dot{\psi}-I_{xy}\ddot{\phi}\cos\psi$

$\frac{\partial T}{\partial\psi}=0+I_{xy}\dot{\psi}\dot{\phi}\sin\psi+0+I_{xx}\dot{\phi}^2\sin\psi\cos\psi=I_{xy}\dot{\psi}\dot{\phi}\sin\psi+I_{xx}\dot{\phi}^2\sin\psi\cos\psi$

$$\Rightarrow M_x=I_{xx}\ddot{\psi}+\cancel{I_{xy}\dot{\phi}\sin\psi\,\dot{\psi}}-I_{xy}\ddot{\phi}\cos\psi-(\cancel{I_{xy}\dot{\psi}\dot{\phi}\sin\psi}+I_{xx}\dot{\phi}^2\sin\psi\cos\psi)$$
$$=I_{xx}\ddot{\psi}-I_{xy}\ddot{\phi}\cos\psi-I_{xx}\dot{\phi}^2\sin\psi\cos\psi$$
$$=I_{xx}\ddot{\psi}-I_{xy}\ddot{\phi}\cos\psi-\frac{1}{2}I_{xx}\dot{\phi}^2\sin(2\psi)$$

Trig. Identity $\sin(2\psi)=2\sin\psi\cos\psi$
$\frac{1}{2}\sin(2\psi)=\sin\psi\cos\psi$

$\therefore$ $\boxed{I_{xx}\ddot{\psi}=M_x+I_{xy}\ddot{\phi}\cos\psi+\frac{1}{2}I_{xx}\dot{\phi}^2\sin(2\psi)}$

Problem 2 1. Turn the system of equations into four first order differential equations $\frac{d\vec{x}}{dt}=g(\vec{x})$.

Given $I\ddot{\phi}+T|\dot{\phi}|\dot{\phi}+\frac{K\phi}{T^2}=\frac{F}{T}$, $\ddot{F}+r_3(1+K)\dot{F}+Kr_3^2F=-\mu Kr_3^2\dot{\phi}$

State vector $\vec{x}=\begin{bmatrix}\phi\\ \dot{\phi}\\ F\\ \dot{F}\end{bmatrix}$ $x_1=\phi$, $x_2=\dot{\phi}$, $x_3=F$, $x_4=\dot{F}$ $\dot{x}_1=x_2$, $\dot{x}_3=x_4$

$\Rightarrow I\ddot{\phi}+T|\dot{\phi}|\dot{\phi}+\frac{K\phi}{T^2}=\frac{F}{T}$ $I\ddot{\phi}=\frac{F}{T}-T|\dot{\phi}|\dot{\phi}-\frac{K\phi}{T^2}$ $\Rightarrow$ $\ddot{\phi}=\frac{1}{I}\left(\frac{F}{T}-T|\dot{\phi}|\dot{\phi}-\frac{K\phi}{T^2}\right)$ $\Rightarrow$ $\dot{x}_2=\frac{1}{I}\left(\frac{x_3}{T}-T|x_2|x_2-\frac{Kx_1}{T^2}\right)$

$\Rightarrow$ $\ddot{F}+r_3(1+K)\dot{F}+Kr_3^2F=-\mu Kr_3^2\dot{\phi}$ $\Rightarrow$ $\ddot{F}=-r_3(1+K)\dot{F}-Kr_3^2F-\mu Kr_3^2\dot{\phi}$ $\Rightarrow$ $\dot{x}_4=-r_3(1+K)x_4-Kr_3^2x_3-\mu Kr_3^2x_2$

$$\dot{\vec{x}}=g(\vec{x})=\begin{bmatrix}x_2\\ \frac{1}{I}\left(\frac{x_3}{T}-T|x_2|x_2-\frac{Kx_1}{T^2}\right)\\ x_4\\ -r_3(1+K)x_4-Kr_3^2x_3-\mu Kr_3^2x_2\end{bmatrix}$$

2. Determine the fixed point(s) of this system, $g(\vec{x}^*)=0$ $\dot{\vec{x}}=0$, $\dot{x}_1=x_2=0$ $x_2^*=0$, $\dot{x}_3=x_4=0$ $x_4^*=0$

$\dot{x}_2=0=\frac{1}{I}\left(\frac{x_3}{T}-T|x_2|\cancel{x_2}^{0}-\frac{Kx_1}{T^2}\right)$ $\Rightarrow$ $0=\frac{x_3}{T}-\frac{Kx_1}{T^2}$ $Tx_3=Kx_1$ $0=Kx_1$ $x_1^*=0$

$\dot{x}_4=0=-r_3(\cancel{1+K})\overset{0}{x_4}-Kr_3^2x_3-\mu K\cancel{r_3^2x_2}^{0}=-Kr_3^2x_3$ $\therefore$ $x_3^*=0$ $\therefore$ the fixed point is $\vec{x}^*=\begin{bmatrix}0\\0\\0\\0\end{bmatrix}$

3. Linearize the system about the fixed point, $\frac{d\vec{x}}{dt}\approx\frac{d}{d\vec{x}}g(\vec{x})|_{\vec{x}=\vec{x}^*}(\vec{x}-\vec{x}^*)\to A\vec{x}$.

$A=\left.\frac{dg}{d\vec{x}}\right|_{\vec{x}=\vec{x}^*}$

$$A = \begin{bmatrix} \frac{\partial g_1}{\partial x_1} & \frac{\partial g_1}{\partial x_2} & \frac{\partial g_1}{\partial x_3} & \frac{\partial g_1}{\partial x_4} \\ \frac{\partial g_2}{\partial x_1} & \frac{\partial g_2}{\partial x_2} & \frac{\partial g_2}{\partial x_3} & \frac{\partial g_2}{\partial x_4} \\ \frac{\partial g_3}{\partial x_1} & \frac{\partial g_3}{\partial x_2} & \frac{\partial g_3}{\partial x_3} & \frac{\partial g_3}{\partial x_4} \\ \frac{\partial g_4}{\partial x_1} & \frac{\partial g_4}{\partial x_2} & \frac{\partial g_4}{\partial x_3} & \frac{\partial g_4}{\partial x_4} \end{bmatrix}_{@\,\vec{x}=\vec{x}^*} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -\frac{K}{IT^2} & 0 & \frac{1}{IT} & 0 \\ 0 & 0 & 0 & 1 \\ 0 & -\mu K r_3^2 & -K r_3^2 & -r_3(1+K) \end{bmatrix}$$

4. Construct the characteristic equation for the matrix A. The characteristic equation results from expanding the equation, $\det(A-\lambda I) = 0$, into a fourth order polynomial in λ with four possible eigenvalue solutions $\lambda_{i=1,\ldots,4}$. (Note I is the identity matrix).

$\det(A-\lambda I) = 0$ Look a python code